

Experiment 12

Question

Perform Perspective Transformation on the Image.

Aim

To perform a perspective transformation on an image using Python and OpenCV by changing the image viewpoint.

Procedure

1. Import the required libraries (cv2 and numpy).
2. Read the input image (dog.jpg).
3. Define four source and four destination points.
4. Compute the perspective transformation matrix.
5. Apply the transformation to the image.
6. Display the original and transformed images.
7. Save the transformed image.

Code

```
import cv2
```

```
import numpy as np
```

```
# Read image
```

```
img = cv2.imread("dog.jpg")
```

```
# Get image size
```

```
rows, cols = img.shape[:2]
```

```
# Source points
pts1 = np.float32([[50,50],[250,50],[50,250],[250,250]])

# Destination points
pts2 = np.float32([[10,100],[220,50],[100,250],[250,220]])

# Perspective transformation
M = cv2.getPerspectiveTransform(pts1, pts2)
result = cv2.warpPerspective(img, M, (cols, rows))

# Display images
cv2.imshow("Original", img)
cv2.imshow("Perspective", result)
```

Input

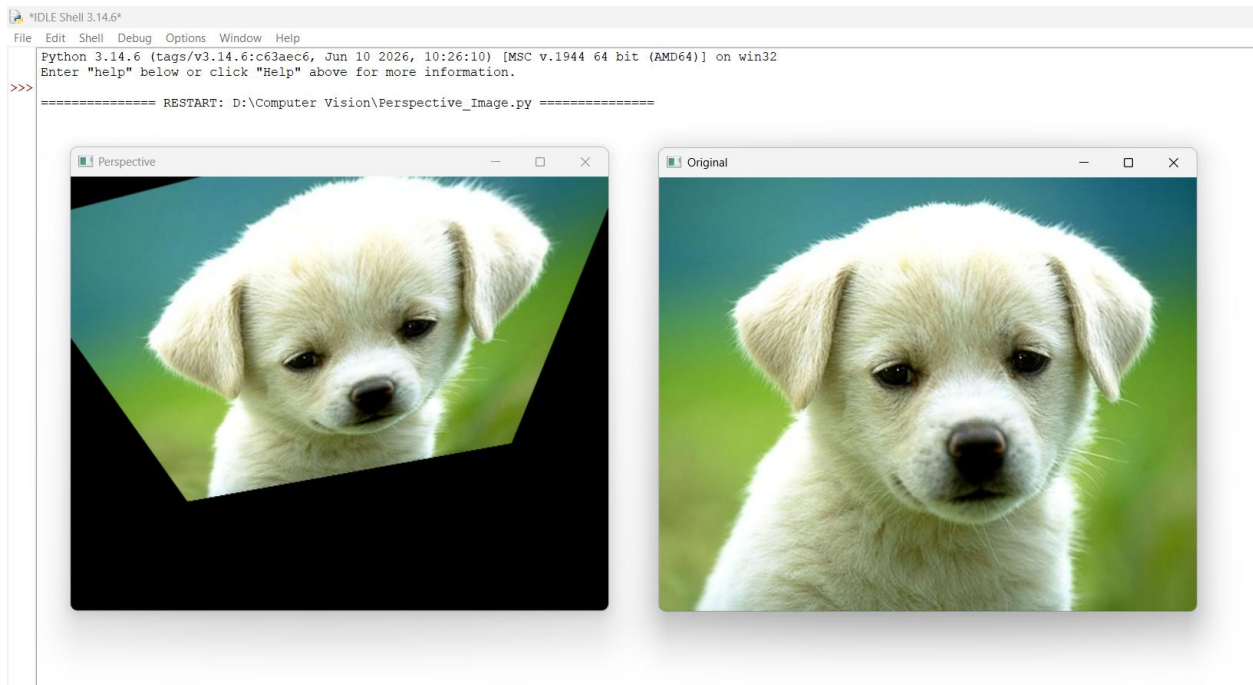
Input Image: dog.jpg

Output

1. Original Image
2. Perspective Transformed Image

Saved file:

- perspective_dog.jpg



Result

The perspective transformation was successfully applied to **dog.jpg**, and the perspective transformed image was displayed and saved successfully.